

3. Differentiation

What students need to learn:

Differentiation of hyperbolic functions and expressions involving them.

For example, $\tanh 3x$, $x \sinh^2 x$, $\frac{\cosh 2x}{\sqrt{x+1}}$.

Differentiation of inverse functions, including trigonometric and hyperbolic functions.

For example, $\arcsin x + x\sqrt{1-x^2}$, $\frac{1}{2} \operatorname{artanh} x^2$.
